



LYCEUM OF THE PHILIPPINES LAGUNA

COLLEGE OF BUSINESS AND ACCOUNTANCY
SECOND SUMMATIVE ASSESSMENT
First Term SY 2026-2027

FM-LPU-L-ACAD-07
Revision: 00
Effective: JAN 2, 2020

BDAT04 (Fundamentals of Data Warehousing)

Course Intended Learning Outcome(s)

This assessment intends to measure attainment of the following:

CILO 2. Profile, clean, transform and combine business data from multiple sources using Power Query, documenting every transformation step so that the result is auditable and repeatable.

CILO 3. Design and build a dimensional model — declaring the grain, specifying fact and dimension tables, keys and relationships — and validate it in Power BI.

CILO	Assessed through	Points
CILO 2, 3	TEST I — Written Examination (situational essays, Questions 1–3, with AI-polish and prompt log)	50
CILO 3	TEST II — Practical Demonstration (four drawn tasks, Bands A–D, on an unseen dataset)	45
CILO 2, 3	TEST III — Oral Interpretation of the Model	5
CILO 3	TEST IV — Bonus Challenge (dimensional modeling vocabulary)	+3

General Instructions:

This midterm examination is administered in two parts within one session and is assessed with Rubric B, re-weighted for this term. Total: 100 points, plus a bonus challenge of up to 3 points. Coverage is Weeks 1 to 6, including the three advance reading packets issued for Weeks 3 and 4 and Session 6. Conditions: open notes for Tests I and II — syllabus, advance readings, feedback letters, laboratory and activity sheets, and the student’s own Power BI model — and NO artificial-intelligence tools of any kind during the examination. Test III is closed-book. Test I is accomplished in a Google Colab notebook distributed through the learning management system; Test II is performed in the Power BI Service.

Prepared by:  Jerald B. Bongalos, MScT, LPT, RChT Course Facilitator	Date Submitted: September 29, 2026	Reviewed and Approved by: Dean / Program Chair
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Test I : Written Examination — Situational Essay (50 points · 90 minutes)

Directions: Answer all three questions inside the examination notebook, between the triple quotes, labeling parts (a), (b), (c). About 250–350 words per answer. The in-room draft carries the substance marks; grammar is not graded on the draft. After the examination, an AI-polished version (grammar and clarity only) and a verbatim prompt log with the shared conversation link are due Sunday, 4 October 2026, 11:59 PM. The draft and polished versions are read side by side; polish that adds ideas, examples, or numbers exceeds its authority.

Test II : Practical Demonstration (45 points · 120 minutes)

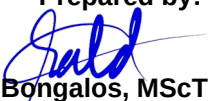
Directions: Each examinee draws by lot FOUR tasks — one from each difficulty band A, B, C and D — and performs them in Power Query on an instructor-prepared dataset not previously seen in the course. The dataset carries THREE fact tables at three different grains, so the tasks examine dimensional design and not cleaning alone. No task depends on a student’s assigned functional case; all tasks sit on the shared competence of Weeks 1–6. Every task ends with a counted EVIDENCE result and a one-sentence written DECISION. Bands and points:

Section	Task drawn (one from each band)	Points
Band A	A single operation with one correct outcome — load, inspect structure and data types, and count what is there	8
Band B	Two steps where the order matters — a repair that fails if performed in the wrong sequence, counted and logged	10
Band C	A diagnostic — find a defect that announces nothing, count it exactly, and state its silent consequence	12
Band D	A judgement — decide between fix, exclude, escalate or accept, and defend the choice in writing	15
—	Total for Test II	45

TEST III: Oral Interpretation of the Model (5 points · ~5 minutes per examinee)

Directions: From their own B3 model, the examinee states the grain of one fact table as a sentence and says why it is that and not something coarser; identifies one relationship and which table sits on the ONE side and which on the MANY, with the reason; and takes one number the model produces and says what it means for the business — not what the formula does. Without notes. The student’s own model may be open on screen; a prepared script may not be read. Assessed per Rubric B’s interpretation criterion.

Criterion	4 — Exemplary	3 — Proficient	2 — Developing	1 — Beginning
1 · Grain and relationship Weight 2.5 States the grain as a sentence and places the relationship correctly, without notes	States the grain as a complete sentence naming every level that varies, and places the ONE and MANY sides correctly with the reason, unprompted.	Grain stated correctly with minor gaps; relationship sides correct and correctly explained.	Grain implied rather than stated; names the sides but cannot say why; needs prompting.	No grain statement, or places the fact on the ONE side; cannot account for the model built.
2 · Business meaning Weight 2.5 Says what the number	Explains what the number means for a named business reader and what it does not establish;	Gives a workable business explanation; may wobble on one probe but recovers by returning to	Explanation restates the formula rather than its meaning; one probe answered, the other met	Cannot interpret the model’s output, or invents a meaning the model does not support.

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Scoring	Level (1–4)	Weight	Points (level ÷ 4 × wt)
Criterion 1 — Grain and relationship		2.5	
Criterion 2 — Business meaning		2.5	
TEST III TOTAL	—	5	/ 5
Bonus card drawn: grain · conformed dimension · degenerate dimension · cardinality (scored separately: +3 / +2 / +1 / 0 — adds only, never deducts)	—	+3 max	

TEST IV: Bonus Challenge (up to +3 points · non-deductive)

One term drawn blind — grain, conformed dimension, degenerate dimension or cardinality. The examinee defines it, says why it matters, and gives one worked example from their own model. Full credit requires all three; a definition on its own earns partial credit. The bonus can only raise a score.

Test I : Written Examination — Situational Essay (50 points · 90 minutes)

Question 1 — The question you could not answer (15 points)

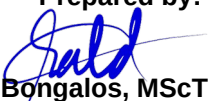
A tricycle operators’ cooperative in Los Baños hires you. They hand you one file, trips.xlsx, with one row for every trip a unit makes: the date, the unit number, the driver, the route, and the total fare collected on that trip. The manager asks two things in the same breath: what is the average number of passengers per trip, and what is the average fare per passenger. A photograph of a full jeepney tells you the jeepney was full; it does not tell you who was on it.

- (a) State the grain of trips.xlsx as a sentence. Then explain which of the two questions the file can answer and which it cannot, using the difference between data that is MISSING and a limitation of the GRAIN. Say clearly which one this is.
- (b) The manager says: “Then just add a passenger column.” Explain, in business terms he would accept, why that does not solve the second question — and what would actually have to change.
- (c) You propose to rebuild. Name the TWO tables you would create, write the grain of each as a full sentence, and name one dimension both would share. Say what that shared dimension lets him do that he cannot do today.

Question 2 — The filter that only moved half the report (15 points)

You have built a model for a hardware chain. One page shows units sold beside units received, month by month, with a date slicer at the top. A branch manager clicks March. The sales bars drop to March. The deliveries bars do not move at all. Nothing turns red and no error appears — the page simply shows March sales against a whole year of deliveries, and it looks like the branch is buried in stock it never sold. Separately, a left anti join from the sales table to the branch list returns forty rows.

- (a) List what must be true of a relationship for a filter to travel from a dimension to a fact — the cardinality, the side each table sits on, and what a marked date table is for. Then name the specific fault behind the unmoving bars, and how you would prove it in one check.
- (b) Explain what a left anti join is actually doing, in your own words. Then distinguish an orphan that is a FORMATTING fault from one that is GENUINELY UNKNOWN, say how you would tell them apart, and how many of the forty you would expect each kind to be — reasoning from your own model.

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- (c) You can repair the dimension, exclude the forty rows, or escalate and report without them. Choose one, and state what the other two would have cost. Argue from this situation, not from a rule.

Question 3 — The model, a year from now (15 points)

Your model works and the CFO likes it. You have been promoted to another department and somebody else now owns the file. Opening it, they find a table called tblSuppliers that no relationship touches, a fact table whose grain you never wrote down anywhere, and a set of applied steps named Changed Type, Changed Type1 and Changed Type2. You can eat a dish once; you can cook from a recipe forever — but a recipe with no measurements is just a list of ingredients.

- (a) Explain ETL in your own words, then say what ELT changes and why it became popular. Using that, state where the transformation lives in a Power Query model like yours — and what that means for the person who inherits it.
- (b) tblSuppliers joins to nothing. Explain why leaving it in the model is a DESIGN FAULT rather than harmless clutter, naming at least one specific thing that can go wrong because it is there. Then say what you would do about it and what you would need to know first.
- (c) Your grain statement currently exists only in your head. Name TWO places it could physically live so that the next person finds it without asking you, and say what specifically goes wrong when it lives nowhere. Use the naming of those applied steps in your answer.

Polish and prompt log (5 points) — assessed on the honesty and quality of the record, per the two-version rule above.

LPU – Laguna student handbook section 7.3.10 states that, "cheating in any form during an examination, written reports, case analysis, experiments and practical demonstrations, or any form of assessment is considered a SERIOUS OFFENSE", and is subject to existing sanctions stated in section 7.4.

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